Technical Audit of Demographic and Intersectional Fairness in ICU Mortality Prediction

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Option B: In-depth Fairness Audit

Background and motivation

ICU mortality scores support clinical risk assessment by estimating a patient's risk of death. Their usefulness depends on how well they identify high-risk patients across populations. We selected this system to study algorithmic accountability, subgroup performance, and the effects of measurement and missing data. Our audit will examine whether mortality risk is systematically underestimated for particular demographic groups and whether overall performance conceals differences in missed high-risk patients.

Existing system, data, and code

We will audit `saminens`'s `WomenInDS.ipynb` solution to the completed WiDS Datathon 2020 competition [3, 2]. The notebook trains LightGBM gradient-boosted trees to estimate `hospital_death` from first-day clinical measurements, diagnoses, and demographic information. Its pipeline includes category consolidation, iterative imputation, categorical encoding, removal of perfectly correlated features, and four-fold stratified validation. The final model excludes gender as a predictor; we retain the original demographic fields for subgroup analysis. The official PhysioNet release describes 91,713 labeled encounters, an unlabeled competition test set, and a data dictionary [1]. Our audit uses the labeled training data downloaded from Kaggle and the cited version of the source implementation.

Research questions

1. How do ROC-AUC, precision, true-positive rate, and false-positive and false-negative rates differ across age, recorded gender, and ethnicity groups?
2. How do false-negative rates and Equal Opportunity differ in detecting patients who subsequently die?
3. What performance differences emerge within age–gender and gender–ethnicity intersections?
4. How do group calibration and these gaps change with the prediction threshold?

Proposed methodology

We will evaluate the source LightGBM approach using a fixed-seed, outcome-stratified split of the labeled encounters, with separate patients in the development and test partitions. We will fit preprocessing on training data and use an inner validation partition for early stopping. A hospital-disjoint evaluation will assess generalization to unseen hospitals. We will document compatibility changes and adaptations to the competition pipeline. Overall evaluation will include ROC-AUC, accuracy, precision, recall, F1, and a confusion matrix.

We will define audit groups using recorded demographics and report missing values as separate groups. Group sizes and death counts will determine which subgroup estimates have sufficient support. Bootstrap confidence intervals and repeated splits will assess the uncertainty and stability of performance gaps. Calibration curves and observed-minus-predicted mortality will measure risk underestimation, while threshold sensitivity will show how missed deaths and false alarms change at different cutoffs. Equal Opportunity will compare detection rates among patients who die; Equalized Odds will also compare false-positive rates among survivors.

Feasibility and expected contribution

We validated the Kaggle training data and ran a single-model adaptation of the source's final LightGBM pipeline. On 18,343 patient-disjoint held-out encounters, ROC-AUC was 0.904; at threshold 0.5, recall was 0.359 and precision was 0.717. Demographic and intersectional evaluation also ran successfully. The preliminary female-minus-male false-negative rate difference was -0.035, with a 95% bootstrap interval of $[-0.087, 0.013]$, motivating further analysis of its stability across splits and thresholds. Our three-person team will jointly extend this proof of concept through calibration analysis, intersectional comparisons, and hospital-disjoint evaluation. We expect to identify where the model performs consistently across groups and where differences in risk estimation or missed deaths warrant closer examination.

References

[1] Meredith Lee, Jesse Raffa, Marzyeh Ghassemi, Tom Pollard, Sharada Kalanidhi, Omar Badawi, Karen Matthys, and Leo Anthony Celi. Wids (women in data science) datathon 2020: Icu mortality prediction. *PhysioNet*, 2020. Version 1.0.0. https://doi.org/10.13026/vc0e-th79.

[2] saminens. Women-in-data-science-2020. https://github.com/saminens/Women-in-Data-Science-2020. Notebook WomenInDS.ipynb; commit `a19d3ba`.

[3] Wids datathon 2020. Kaggle. https://www.kaggle.com/competitions/widsdatathon2020.

AI Disclosure

I used ChatGPT to brainstorm potential research topics and identify existing systems that interested me. I then evaluated the feasibility of each option and tested the selected model locally to confirm that it could be implemented. I also used ChatGPT to help plan the LaTeX layout and formatting of the proposal, but all the written content was my own work.